

SAP Technical Architecture Blueprint

Custom Flexible Workflow for SAP PPM Decision Points via SAP BTP Process Automation

Scope: SAP S/4HANA Private Cloud & SAP BTP | Pattern: Side-by-Side Extensibility & Unified Inbox | Date: September 2026

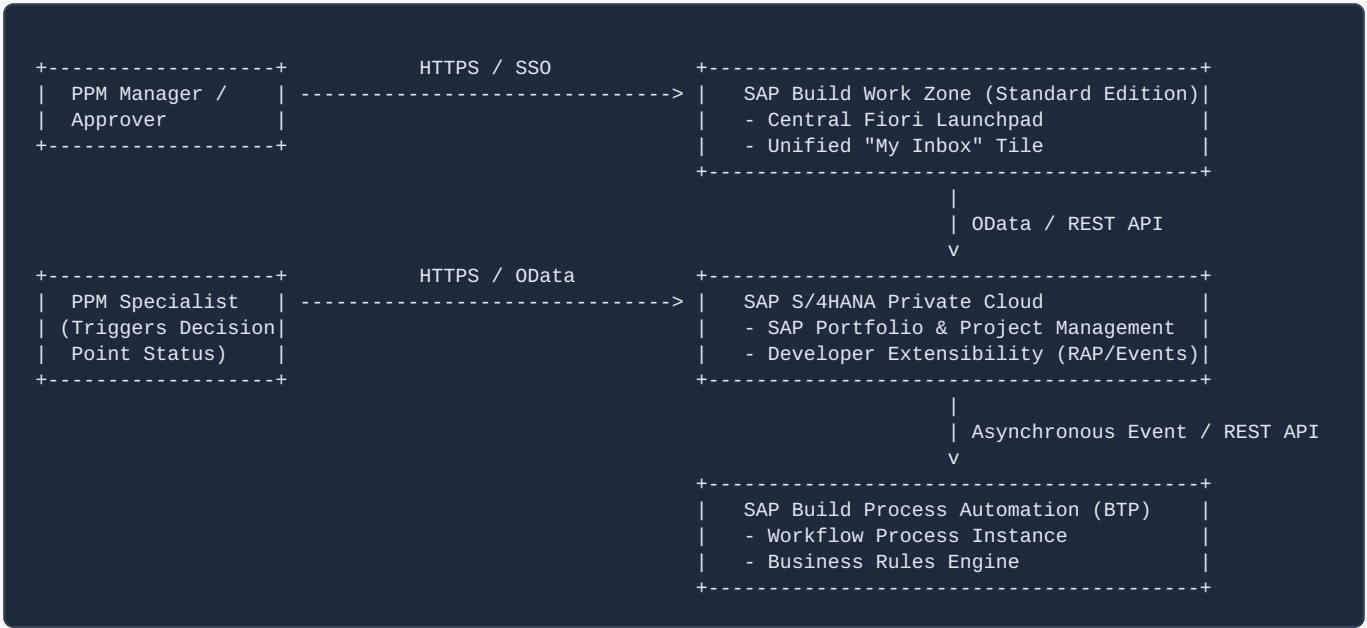
1. Executive Summary & Core Architecture Highlights

This document details the target C4 model architecture for implementing custom flexible approval workflows for **SAP Portfolio and Project Management (PPM) Decision Points**. The business logic and orchestration are completely offloaded to **SAP BTP Process Automation**, ensuring a pristine S/4HANA core while providing business approvers with a single entry point through **SAP Build Work Zone**.

1. Clean Core Strategy Business rules & approvals reside entirely on SAP BTP. Backend updates utilize standard, released RAP/OData interfaces without core modifications.	2. Centralized User Experience Approvers access a unified "My Inbox" tile in SAP Build Work Zone, powered by SAP BTP Task Center / Task Services.	3. Identity & Governance Federated SSO via SAP IAS. Secure backend transaction execution enforced via Cloud Connector Principal Propagation.
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2. C4 Level 1: System Context Diagram

High-level view of system boundaries, primary roles, and core communication paths.



3. C4 Level 2: Container Diagram (System & Network Topology)

Detailed architectural decomposition across Cloud Foundry, Security Infrastructure, and On-Premise/Private Cloud layers.

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USER LAYER

[PPM Approver] ---> Accesses central Launchpad via Web Browser / Mobile

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| HTTPS / SAML 2.0 / OAuth 2.0
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SECURITY & IDENTITY LAYER (SAP BTP)

[SAP IAS (Identity Authentication Service)] ---> Federation with Corporate IdP (Azure AD / Okta)

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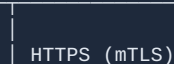
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SAP BTP SUBACCOUNT (Cloud Foundry Environment)

1. SAP Build Work Zone (Unified Shell / Launchpad)
 - Unified My Inbox App (Fetches user-specific tasks from BTP Task Service)



2. SAP Build Process Automation
 - Workflow Trigger: Event-driven / OData Inbound API
 - Decision Logic: SAP Build Process Automation Decisions (Business Rules for approval levels)
 - User Tasks: Form-based Approval Tasks pushed to Task Service
 - Actions / Outbound Call: Destination Service to trigger backend status update



3. SAP BTP Connectivity Services
 - SAP BTP Destination Service (Stores S/4HANA connection details & Principal Propagation auth)
 - SAP Cloud Connector (Secure TLS Tunnel bridging BTP Cloud to S/4HANA Private Network)



↓ Secure Tunnel / Principal Propagation
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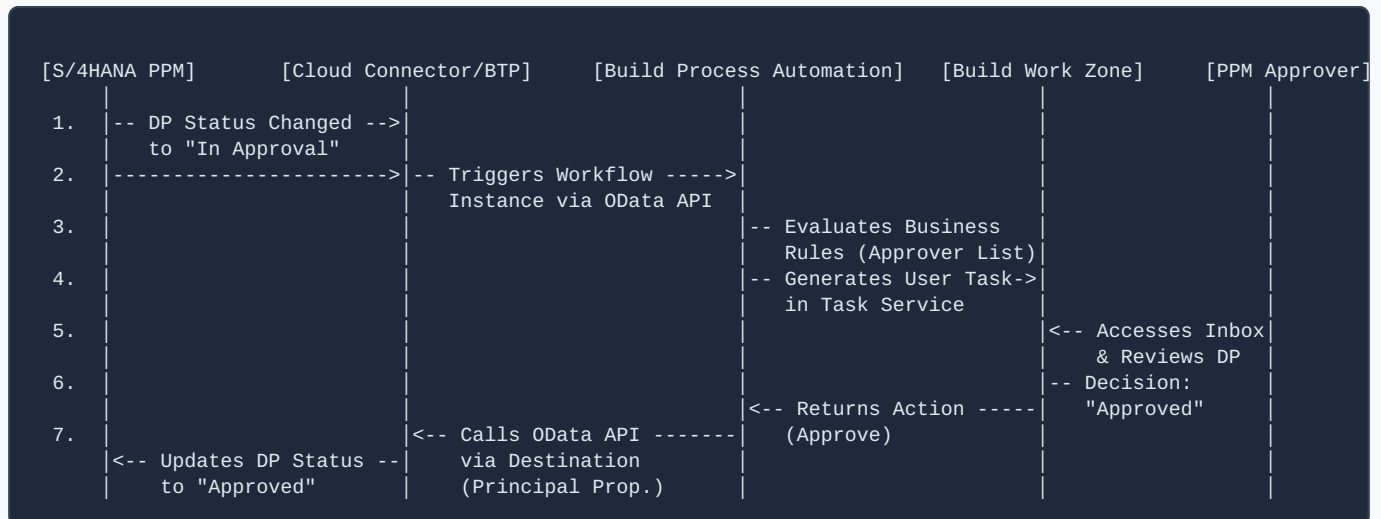
SAP S/4HANA PRIVATE CLOUD BACKEND

1. SAP PPM (Portfolio & Project Management) Module
 - Decision Point (DP) Status Change Listener / Custom BAdI
 - Custom Event Generation (SAP Business Event Handling / Event Mesh / Outbound Webhook)



2. Clean Core Integration Layer (Developer Extensibility / RAP)
 - Released OData v4 Service / RAP B0 (Updates Decision Point status e.g., "Approved", "Rejected")
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4. C4 Level 3: Component Execution Sequence



5. Architectural Decision Summary

Feature Domain	Design Choice	Technical Justification
Workflow Engine	SAP Build Process Automation	Decouples process logic from core S/4HANA, fully aligning with Clean Core guidelines while providing low-code flexibility and central audit logs.
User Experience	SAP Build Work Zone	Provides a single entry point for all enterprise approvals, aggregating tasks across BTP and backend ERP modules via Task Center.
Backend API Layer	RAP / OData v4	Ensures all status updates on PPM Decision Points utilize released, supportable APIs with transactional guarantees.
Security & Identity	IAS + Principal Prop.	Guarantees that the approver's identity in BTP is propagated directly to S/4HANA, ensuring backend authorization checks are enforced.